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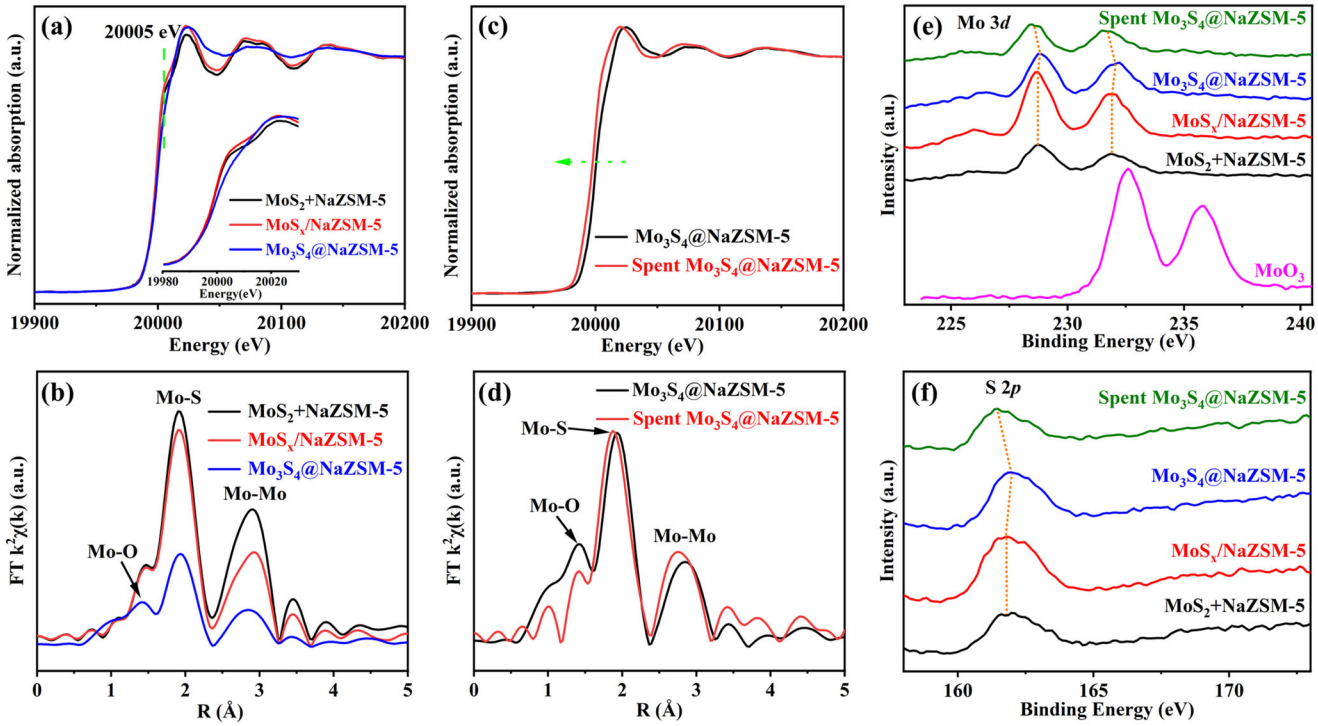

Fig. 3 | Structure and physical property of catalysts. a X-ray absorption near edge structure (XANES) spectroscopy and **b** radical distribution functions obtained by Fourier-transformed extended X-ray absorption fine structure (EXAFS) spectra of commercial MoS_2, MoS_x/NaZSM-5, and Mo_3S_4@NaZSM-5; **c** XANES spectroscopy and **d** radical distribution functions obtained by Fourier-transformed EXAFS spectra of Mo_3S_4@NaZSM-5 and spent sample; **e** Mo 3*d* XPS spectra of MoO_3, commercial MoS_2 + NaZSM-5, MoS_x/NaZSM-5, Mo_3S_4@NaZSM-5, and Spent Mo_3S_4@NaZSM-5; **f** S 2*p* XPS spectra of commercial MoS_2 + NaZSM-5, MoS_x/NaZSM-5, Mo_3S_4@NaZSM-5, and Spent Mo_3S_4@NaZSM-5.

amount of molybdenum added in synthesis. Compared with NaZSM-5 and MoS_x/NaZSM-5, Mo_3S_4@NaZSM-5 has the lowest Brunauer-Emmett-Teller (BET) specific surface area (Supplementary Table 2), indicating that $[Mo_3S_4]^{n+}$ clusters occupy some zeolitic pores of ZSM-5 zeolite. While little difference is found between the specific surface area of MoS_x/NaZSM-5 and NaZSM-5, consistent with the external loading of MoS_2 on NaZSM-5 zeolite.

The confinement of metallic centers for hydrogenation by the zeolitic framework in catalyst Mo_3S_4@ZSM-5 is further checked by shape-selective hydrogenations of 2,3-dimethylnitrobenzene and nitrobenzene[19,27–31] and the results are listed in Supplementary Fig. 4 and Supplementary Table 3, indicating that $[Mo_3S_4]^{n+}$ is located in the interior of ZSM-5 zeolite for Mo_3S_4@ZSM-5, different from that of MoS_x/ZSM-5.

Catalytic hydrogenation of CO_2

The catalytic properties of Mo_3S_4@NaZSM-5 catalyst for CO_2 hydrogenation are measured and depicted in Fig. 4. As shown in Fig. 4a, the conversion of CO_2 increases with the increase of reaction temperature from 120 °C to 270 °C. When the reaction temperature raises from 150 to 180 °C, a conversion jump can be seen, because of the desorption of product methanol from metallic centers is limited by the peripheral NaZSM-5 framework at lower temperatures, as revealed by results of methanol-TPD (Supplementary Fig. 5). At 180 °C, a selectivity to CH_3OH larger than 98% is obtained at CO_2 conversion of 10.2% and the selectivity to CO + CH_4 is less than 2%. When the reaction temperature exceeds 180 °C, the increase in amount of CO and CH_4 lowers the selectivity to CH_3OH, driven by thermodynamics (Supplementary Fig. 1). With the pressure rising, as described in Fig. 4b, both the conversion of CO_2 and the selectivity of CH_3OH increase, due to the principle of Le Chatelier. Noteworthy, the effect of space velocity on the reaction is very interesting and it can be seen from the Fig. 4c that the CH_3OH selectivity increases and the selectivity to CO + CH_4 decreases with the space velocity increasing. It is a little strange but well elucidated by DFT calculations, which clearly show that the energy barrier of *CO desorption is much higher than hydrogenation of *CO to *CHO (see below).

The service life of catalyst is of great importance for industrialization. A fatal problem of the hydrogenation of CO_2 reported to date is that the deactivation of the most of supported metal catalysts due to the sintering of active components at high temperatures[32]. For current Mo_3S_4@NaZSM-5 catalyst, the active component $[Mo_3S_4]^{n+}$ is protected by the surrounding ZSM-5 zeolitic framework which is extremely stable at relatively low reaction temperature of 180 °C. It keeps catalytic performance, i.e., CO_2 conversion >10% and at the same time the CH_3OH selectivity >98%, without any reduction in continuous 1000 h on stream of CO_2 hydrogenation (Fig. 4d, 180 °C, 4 MPa, 1200 mL g_{cat}^{-1} h^{-1}).

For comparison, the catalysts MoS_x/NaZSM-5 and commercial MoS_2 are also tested for the reaction and the specific experimental data are listed in Table 2. Interestingly, both of them

Table 1 | Curve-fitting results for various samples obtained by Fourier-transformed EXAFS spectra

Samples	Mo–S			Mo–Mo		
	N[a]	R (Å)[b]	σ²(Å²)[c]	N[a]	R (Å)[b]	σ²(Å²)[c]
Mo_3S_4@NaZSM-5	3.0 ± 0.2	2.332	0.0025	1.9 ± 0.4	2.931	0.0042
Spent Mo_3S_4@NaZSM-5	3.0 ± 0.3	2.310	0.0029	2.0 ± 0.3	2.93	0.0036
MoS_x/NaZSM-5	5.7 ± 0.3	2.321	0.0027	4.8 ± 0.6	3.152	0.0045
MoS_2 + NaZSM-5	6	2.321	0.0035	6	3.150	0.0039

[a]Coordination numbers.
[b]Bond distances.
[c]Debye-Waller factor.